

# Solution Blueprint: Aerospace Repair Engineering Order (REO) Compliance Review Agent

## Automated Airworthiness Audit for Commercial Aircraft Fleet Support

Hybrid Cognitive Architecture Combining Deterministic Engines & Multimodal LLMs on Google Cloud

### Executive Document Information

| Attribute                | Specification                                                                                                                        |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Client / Domain:         | Commercial Aircraft Manufacturing & Engineering Fleet Support                                                                        |
| Solution Category:       | Airworthiness Compliance Automation (Repair Engineering Orders – REOs)                                                               |
| Technology Stack:        | Google Agent Development Kit (ADK) 2.0, Gemini 3.7 Flash Multimodal, Vertex AI Agent Engine, Google Cloud Storage, Gemini Enterprise |
| Hosting Region:          | Google Cloud <code>europa-west1</code> (Belgium) / Runtime: Vertex AI Reasoning Engine                                               |
| Document Classification: | Enterprise Solution Blueprint & Validated Reference Architecture                                                                     |
| Status:                  | Active in Production / Validated Reference Architecture                                                                              |
| Version:                 | 1.0.0 (Anonymized Enterprise Distribution)                                                                                           |

### Executive Summary & Solution Highlights

**Google Cloud Architecture Statement:**

The **Aerospace REO Compliance Review Agent** is an enterprise AI solution designed to automate and accelerate the airworthiness compliance audit of Repair Engineering Orders (REOs). In commercial aircraft maintenance and fleet support, REOs are mission-critical engineering authorizations that specify permanent or temporary structural repair dispositions for damaged aircraft components.

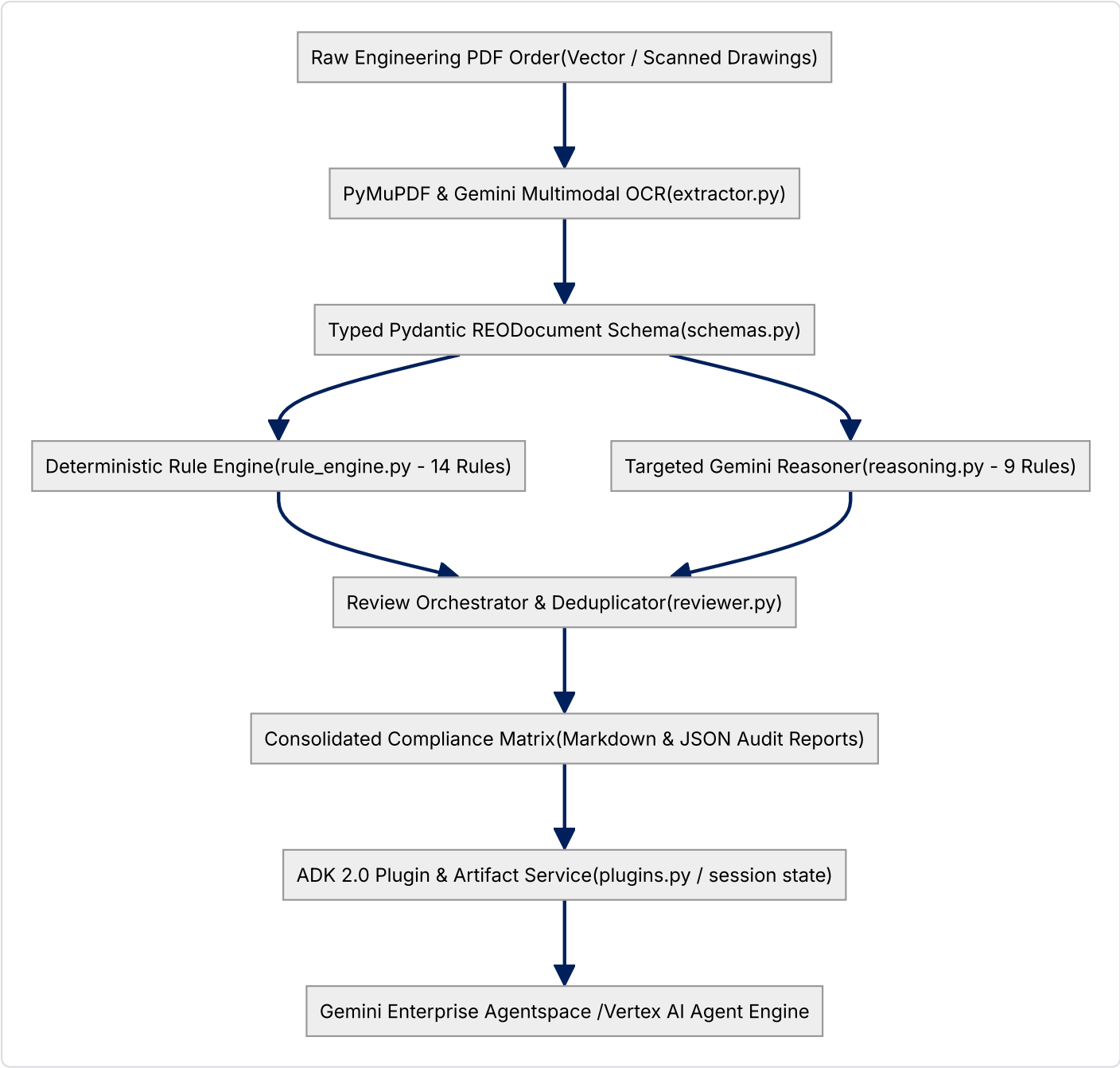
Every REO document must satisfy a rigorous **24-point engineering compliance checklist** covering structural effectivity, title syntax, drawing orientation, SDIR tables, and imperative disposition formatting. Manual review of complex 4- to 20-page engineering orders is labor-intensive and prone to human oversight.

This solution delivers a **hybrid cognitive architecture**:

- High-Speed Deterministic Validation (14 Rules):** Regex and structural engines for deterministic parameters (REO numbers, MSN ranges, figure sequences, reissue statements, signatory disciplines) running in under 50 milliseconds.
- Targeted Multimodal LLM Reasoning (9 Rules):** Gemini 3.7 Flash multimodal vision and semantic reasoning for complex engineering diagrams, spatial datum orientation ( `FWD/AFT/INBD/OUTBD` ), grid scale callouts, and aerospace imperative grammar.
- ADK 2.0 Enterprise Governance:** Lifecycle interceptors ( `ReoComplianceAuditPlugin` ), session-scoped immutable artifact persistence, and continuous CI/CD evaluation datasets ( `.evalset.json` ).

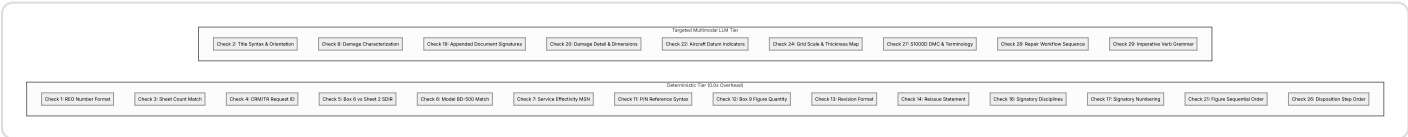
| ⚡ Sub-Second Deterministic Rules       | 📡 Multimodal Visual Precision                   | 🔒 EASA/FAA Part 21 Auditability              | 🌐 Agentspace Integration                        |
|----------------------------------------|-------------------------------------------------|----------------------------------------------|-------------------------------------------------|
| 48 ms Latency for 14 structural checks | Gemini 3.7 Flash Vision Datum & grid inspection | ADK 2.0 Interceptors Immutable session state | Gemini Enterprise & CLI Automated approval flow |

1. End-to-End Hybrid Cognitive Architecture



2. 24-Point Compliance Checklist Architecture

The agent partitions the engineering compliance checklist into two complementary execution tiers:



## Detailed Compliance Matrix Mapping

| #  | Checklist Rule                    | Category           | Execution Engine | Verification Logic                                                                                                                              |
|----|-----------------------------------|--------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | REO # Format & Header Consistency | Cover / Header     | Deterministic    | Regex <code>^500-\d{2}-\d{2}-\d{3,5}\$</code> validated across Box 2 and all page headers.                                                      |
| 2  | Box 3 Title Format & Consistency  | Cover Sheet        | LLM Reasoner     | Validates <code>[REPAIR/DISPOSITION] FOR [DAMAGE] TO [SIDE] [COMPONENT] AT [LOCATION]</code> .                                                  |
| 3  | Box 4 Sheet Count vs Pages        | Cover / Pagination | Deterministic    | Matches <code>1 OF N</code> declaration against physical extracted page count.                                                                  |
| 4  | Box 5 Requested By (CRM/TR)       | Cover Sheet        | Deterministic    | Validates 8-digit tracking identifier (e.g., <code>TR 81845789</code> ), <code>N/A</code> , or delimited list.                                  |
| 5  | Box 6 Limitations vs Sheet 2 SDIR | Cover / SDIR       | Deterministic    | Verifies <code>YES, SEE SHEET 2</code> with active SDIR rows vs <code>NONE</code> with empty SDIR.                                              |
| 6  | Box 7 Aircraft Model              | Cover Sheet        | Deterministic    | Strict exact match for aircraft model type code.                                                                                                |
| 7  | Box 8 Service Effectivity         | Cover Sheet        | Deterministic    | Validates non-serialized MSNs ( <code>50010-54999</code> ) or serialized <code>P/N, S/N</code> .                                                |
| 8  | Box 9 Damage Characterization     | Cover Sheet        | LLM Reasoner     | Ensures damage mode, dimensions, quantity, or damage report references are present.                                                             |
| 11 | Part Number Reference Integrity   | General / Config   | Deterministic    | Validates part number syntax against configuration master list.                                                                                 |
| 12 | Box 9 Figure Quantity Match       | Cover / Figures    | Deterministic    | Compares declared <code>FIGURES 1 TO N</code> in Box 9 against physical figure count.                                                           |
| 13 | Box 10 Revision Format            | Cover Sheet        | Deterministic    | Validates initial <code>--</code> , single letters ( <code>-A</code> , <code>-B</code> ), and prohibits letters <code>I, O, Q, S, X, Z</code> . |
| 14 | Revision Consistency & Reissue    | Cover / Header     | Deterministic    | Verifies revision headers; requires <code>THIS REO IS COMPLETELY RE-ISSUED AT REV XX.</code>                                                    |
| 16 | Engineering Signatory Functions   | Signatures         | Deterministic    | Validates engineering disciplines against authority matrix.                                                                                     |
| 17 | Signatory Numbering (15A, 15B)    | Signatures         | Deterministic    | Enforces sequential alphabetical suffixing ( <code>15A</code> , <code>15B</code> , <code>15C</code> ) for multi-signatories.                    |
| 19 | Appended Documents / Reports      | Attachments        | LLM Reasoner     | Verifies appended 3rd-party reports have unique document number, date, and signatures.                                                          |
| 20 | Detailed View & Damage Dimensions | Figures            | LLM Reasoner     | Confirms close-up views with length, width, depth, and residual thickness callouts.                                                             |
| 21 | Figures Sequential Numbering      | Figures            | Deterministic    | Ensures figures follow strict sequential numbering ( <code>Figure 1, Figure 2...</code> ) without gaps.                                         |
| 22 | General View & Datum Orientation  | Figures            | LLM Reasoner     | Checks for aircraft datum coordinates ( <code>STA, WL, BL, Rib, Frame</code> ) and directional arrows ( <code>FWD, AFT</code> ).                |
| 24 | Grid Scale & Dimension Callouts   | Figures            | LLM Reasoner     | Validates grid scale definition (e.g., <code>10x10 mm</code> ), residual thickness maps, and units.                                             |
| 26 | Disposition Step Numbering        | Disposition        | Deterministic    | Validates sequential paragraph numbering ( <code>1, 2, 3...</code> ) and sub-steps ( <code>A, B, C...</code> ).                                 |
| 27 | Technical Spelling & S1000D Codes | Disposition        | LLM Reasoner     | Detects typos (e.g., <code>NUPLATE → NUTPLATE</code> ) and validates S1000D DMC codes.                                                          |
| 28 | Disposition Step Sequencing       | Disposition        | LLM Reasoner     | Verifies logical repair workflow (blend → NDT → fastener install → corrosion inhibitor).                                                        |
| 29 | Imperative Verb Phrasing          | Disposition        | LLM Reasoner     | Enforces uppercase imperative action verbs ( <code>REMOVE</code> , <code>PERFORM</code> , <code>REPLACE</code> , <code>WET INSTALL</code> ).    |

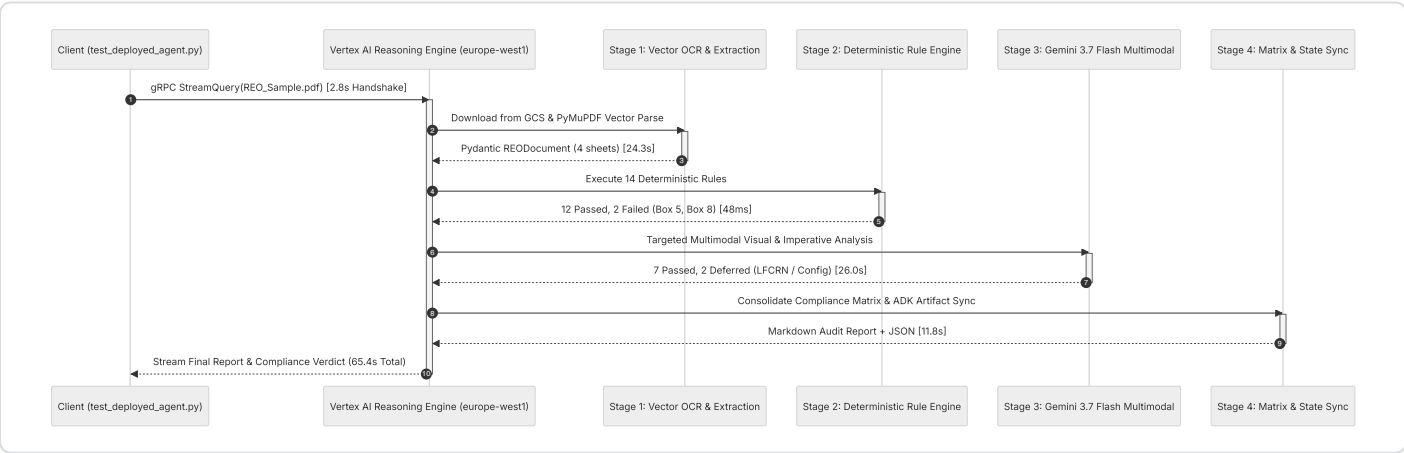
### 3. Live Execution Telemetry & Performance Trace

The deployed agent was benchmarked live on Google Cloud Vertex AI Reasoning Engine via gRPC streaming telemetry ( `scripts/test_deployed_agent.py` ):

| AEROSPACE REO AGENT - TEST SAMPLE AUDIT EVALUATION SUMMARY |                |     |     |      |      |      |      |        |  |  |
|------------------------------------------------------------|----------------|-----|-----|------|------|------|------|--------|--|--|
| Sample Document                                            | REO #          | Rev | Pgs | Pass | Fail | Warn | TODD | Status |  |  |
| REO-500-53-21-1664_-- (Anonymized).pdf                     | 500-53-21-1664 | --  | 19  | 17   | 3    | 1    | 2    | ✗ FAIL |  |  |
| REO-500-53-21-691_-A (Anonymized).pdf                      | 500-53-21-691  | -A  | 7   | 18   | 2    | 1    | 2    | ✗ FAIL |  |  |
| REO-500-57-51-753_-- (Anonymized).pdf                      | 500-57-51-753  | --  | 4   | 19   | 2    | 0    | 2    | ✗ FAIL |  |  |

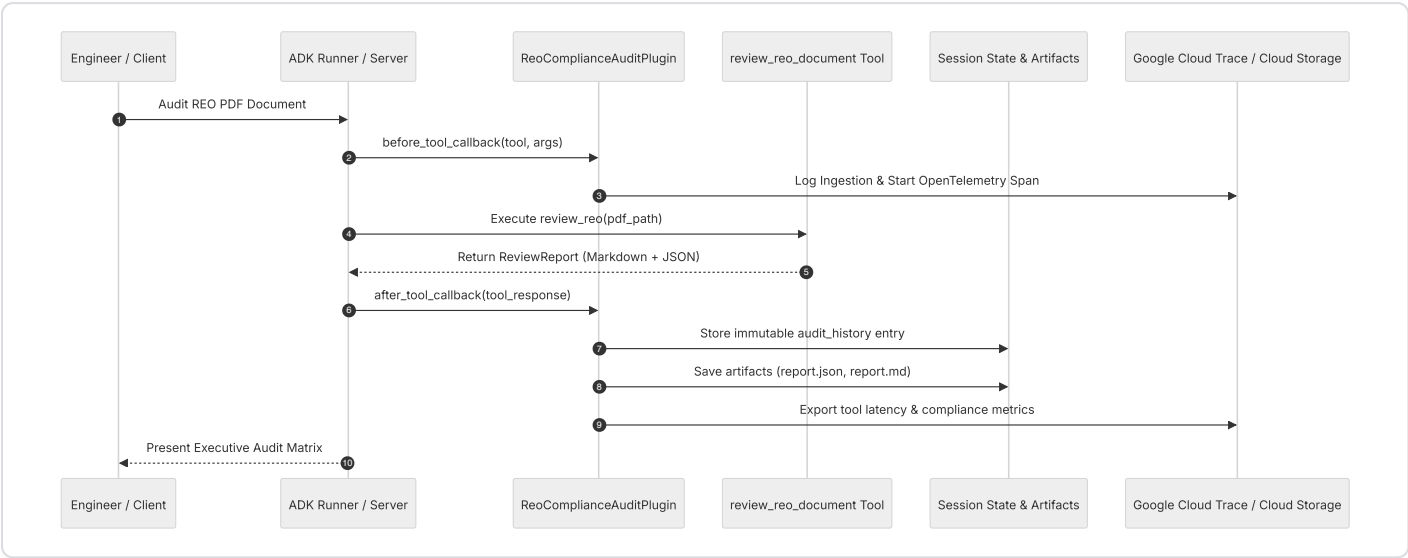
#### Latency Breakdown by Phase (4-Page Engineering Order)

| Phase   | Pipeline Component                                   | Latency | % of Total | Operational Description                                                   |
|---------|------------------------------------------------------|---------|------------|---------------------------------------------------------------------------|
| Phase 1 | gRPC Stream Connect & Session Handshake              | 2.8s    | 4.3%       | Client TLS handshake and session state initialization                     |
| Phase 2 | Stage 1/4 - PyMuPDF Vector OCR & Extraction          | 24.3s   | 37.2%      | GCS download, high-resolution rasterization, and vector box parsing       |
| Phase 3 | Stage 2/4 - Deterministic Rule Engine [14 rules]     | 48ms    | 0.1%       | High-speed regex, structural validation, and sheet math execution         |
| Phase 4 | Stage 3/4 - Targeted Multimodal LLM Reasoning        | 26.0s   | 39.8%      | Gemini 3.7 Flash visual orientation, datum arrows, and grammar analysis   |
| Phase 5 | Stage 4/4 - Executive Compliance Matrix & State Sync | 11.8s   | 18.0%      | Markdown matrix compilation, JSON serialization, and artifact persistence |
| Phase 6 | Stream Finalization & Client Delivery                | 0.4s    | 0.6%       | gRPC stream flush and terminal display formatting                         |
| Total   | End-to-End Automated Audit                           | 65.4s   | 100%       | Full engineering compliance audit completed                               |



## 4. ADK 2.0 Governance & Audit Architecture

To meet aerospace regulatory airworthiness requirements (EASA / FAA Part 21), the agent incorporates ADK 2.0 enterprise governance features:



### Key Governance Invariants

- Lifecycle Interceptors ( `ReoComplianceAuditPlugin` )**: The plugin intercepts every tool invocation. `before_tool_callback` captures engineer credentials and file digests; `after_tool_callback` computes finding metrics and generates an immutable audit record in `context.state["audit_history"]`.
- Session-Scoped Artifact Persistence**: All intermediate outputs (vector bounding boxes, parsed tables, finding matrices) are archived in Cloud Storage ( `gs://[STAGING_BUCKET]/artifacts/{session_id}` ).
- CI/CD Regression Sets**: Evaluation datasets ( `eval_set.evalset.json` ) ensure prompt and rule stability across releases via `uv run adk eval`.

## 5. Deployment & Production Operations

### Cloud Runtime Configuration

- Platform**: Google Cloud Vertex AI Agent Engine (Reasoning Engine)
- Container Region**: `europe-west1` (Low-latency EU sovereign runtime)
- Model Endpoint**: `gemini-3.7-flash` (Vertex AI Global / Regional Endpoint)
- Agentspace Integration**: Discovery Engine Agentspace Assistant registration

```
# Verify local compliance test suite
uv run python scripts/test_samples.py

# Deploy to Vertex AI Agent Engine
uv run python scripts/deploy_agent.py --region europe-west1

# Register agent with Gemini Enterprise Agentspace
bash scripts/register_agent.sh --app-id [AGENTSPEACE_APP_ID]
```